

Roll No.

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Q.P. Code

82/5/1

Candidates must write the Q.P. Code on the title page of the answer-book.

NOTE

- (I) Please check that this question paper contains printed pages.
- (II) Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- (III) Please check that this question paper contains the full number of questions.
- (IV) Please write down the serial number of the question in the answer-book before attempting it.
- (V) 15 minute time has been allotted to read this question paper. The question paper will be distributed 15 minutes prior to the commencement of the examination.

MATHEMATICS (STANDARD)

(CBSE Board Examination — 2025-26)

Time allowed : 3 hours**Maximum Marks : 80****General Instructions :**

- (i) This Question Paper has 5 Sections A, B, C, D, and E.
- (ii) All Sections are compulsory.
- (iii) Section A has 20 MCQs of 1 mark each.
- (iv) Section B has 5 questions of 2 marks each.
- (v) Section C has 6 questions of 3 marks each.
- (vi) Section D has 4 questions of 5 marks each.
- (vii) Section E has 3 Case Study Based Questions of 4 marks each.
- (viii) Internal choices are provided in 2 questions in Section B, 2 questions in Section C, 2 questions in Section D, and in 1 sub-part of each of the 3 case studies in Section E. You have to attempt only one of the alternatives in all such questions.
- (ix) Draw neat figures wherever required. Take $\pi = 22/7$ wherever required, unless stated otherwise.
- (x) Use of calculators is NOT permitted.

SECTION AThis section comprises Multiple Choice Questions (MCQs) of 1 mark each. **20 x 1 = 20**

1. If $\text{HCF}(90, 144) = 18$, then $\text{LCM}(90, 144)$ is: **1**
- (a) 720 (b) 360
(c) 1440 (d) 12960
2. If one zero of the polynomial $f(x) = (k^2 + 4)x^2 + 13x + 4k$ is the reciprocal of the other, then the value of k is: **1**
- (a) 2 (b) -2
(c) 1 (d) -1
3. For what value of 'k' do the equations $kx - y = 2$ and $6x - 2y = 3$ have a unique solution? **1**
- (a) $k = 3$ (b) $k \neq 3$
(c) $k = 0$ (d) $k \neq 0$

4. Find the value of 'p' for which the quadratic equation $px(x-3) + 9 = 0$ has real and equal roots. 1
 (a) 3 (b) 4
 (c) 9 (d) 12
5. The common difference of an AP whose nth term is given by $a_n = 5n - 7$ is: 1
 (a) 5 (b) -7
 (c) -2 (d) 12
6. If $\triangle ABC \sim \triangle PQR$, $\text{area}(\triangle ABC) = 81 \text{ cm}^2$ and $\text{area}(\triangle PQR) = 144 \text{ cm}^2$. If $QR = 6 \text{ cm}$, then BC is equal to: 1
 (a) 9 cm (b) 4.5 cm
 (c) 3 cm (d) 8 cm
7. The distance of the point $P(4, -3)$ from the origin is: 1
 (a) 7 units (b) 5 units
 (c) 1 unit (d) $\sqrt{7}$ units
8. The value of $(\sin 30^\circ + \cos 60^\circ) - \tan 45^\circ$ is: 1
 (a) 0 (b) 1
 (c) $1/2$ (d) $\sqrt{3}/2$
9. If $\sec A + \tan A = x$, then $\sec A - \tan A$ is: 1
 (a) $1/x$ (b) x
 (c) x^2 (d) 1
10. In the given figure, if two tangents TP and TQ are drawn to a circle with centre O such that $\angle POQ = 110^\circ$, then $\angle PTQ$ is equal to: 1

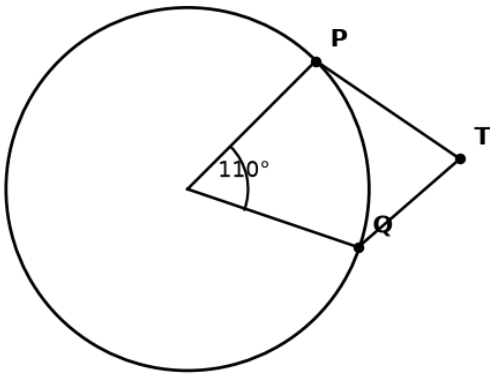


Fig. 1

- (a) 60° (b) 70°
 (c) 80° (d) 90°
11. The area of a sector of a circle with radius 6 cm and angle 60° is: 1
 (a) 18.84 cm^2 (b) 37.68 cm^2
 (c) $6\pi \text{ cm}^2$ (d) $12\pi \text{ cm}^2$

- 12.** A metallic sphere of radius 3 cm is melted and recast into a cylinder of radius 2 cm. The height of the cylinder is: **1**
- (a) 9 cm (b) 12 cm
(c) 18 cm (d) 27 cm
- 13.** If the mean and median of a data are 16 and 18 respectively, then the mode of the data is: **1**
- (a) 20 (b) 22
(c) 24 (d) 27
- 14.** A card is drawn at random from a well-shuffled deck of 52 playing cards. The probability of drawing a black face card is: **1**
- (a) $\frac{1}{26}$ (b) $\frac{3}{26}$
(c) $\frac{3}{13}$ (d) $\frac{1}{13}$
- 15.** The angle of elevation of the top of a tower from a point on the ground, which is 30m away from the foot of the tower, is 30° . The height of the tower is: **1**
- (a) $30\sqrt{3}$ m (b) $10\sqrt{3}$ m
(c) $15\sqrt{3}$ m (d) 30 m
- 16.** The sum of the first 'n' positive odd integers is: **1**
- (a) n (b) n^2
(c) $n(n+1)$ (d) $\frac{(n+1)}{2}$
- 17.** Assertion (A): The number $(2 + \sqrt{3})$ is irrational. **1**
Reason (R): The sum of a rational number and an irrational number is always irrational.
- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true but Reason (R) is false. (d) Assertion (A) is false but Reason (R) is true.
- 18.** Assertion (A): If $\triangle ABC$ and $\triangle DEF$ are similar triangles such that $\angle A = 47^\circ$ and $\angle E = 83^\circ$, then $\angle C = 50^\circ$. **1**
Reason (R): The sum of the angles in a triangle is 180° .
- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true but Reason (R) is false. (d) Assertion (A) is false but Reason (R) is true.
- 19.** Assertion (A): The length of the tangent from an external point P to a circle with centre O and radius 5 cm is 12 cm. The distance of P from O is 13 cm. **1**
Reason (R): The tangent at any point of a circle is perpendicular to the radius through the point of contact.
- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A). (b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true but Reason (R) is false. (d) Assertion (A) is false but Reason (R) is true.

20. Assertion (A): If the roots of the quadratic equation $x^2 + 2kx + (k^2 - 5) = 0$ are real and distinct, then $k > \sqrt{5}$ or $k < -\sqrt{5}$. 1
Reason (R): A quadratic equation $ax^2 + bx + c = 0$ has real and distinct roots if its discriminant $D > 0$.
- (a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(b) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(c) Assertion (A) is true but Reason (R) is false.
(d) Assertion (A) is false but Reason (R) is true.

SECTION B

This section comprises Very Short Answer (VSA) Type Questions of 2 marks each. **5 x 2 = 10**

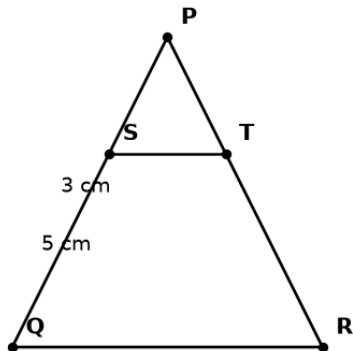
SECTION C

This section comprises Short Answer (SA) Type Questions of 3 marks each. **6 x 3 = 18**

26. Prove that $(3 + 2\sqrt{5})$ is an irrational number, given that $\sqrt{5}$ is an irrational number. 3
27. The sum of the first 7 terms of an Arithmetic Progression (AP) is 49 and that of 17 terms is 289. Find the sum of its first 'n' terms. 3
28. If the points $A(x, y)$, $B(1, 2)$ and $C(7, 0)$ are collinear, show that $x + 3y = 7$. 3

OR

In $\triangle PQR$, S and T are points on sides PQ and PR respectively such that $ST \parallel QR$. If $PS = 3$ cm, $SQ = 5$ cm and $PR = 8$ cm, find the length of PT.



29. Prove the identity: $\frac{\sin \theta - \cos \theta + 1}{\sin \theta + \cos \theta - 1} = \frac{1}{\sec \theta - \tan \theta}$ 3

30. Two tangents TP and TQ are drawn to a circle with centre O from an external point T. Prove that $\angle PTQ = 2\angle OPQ$. 3

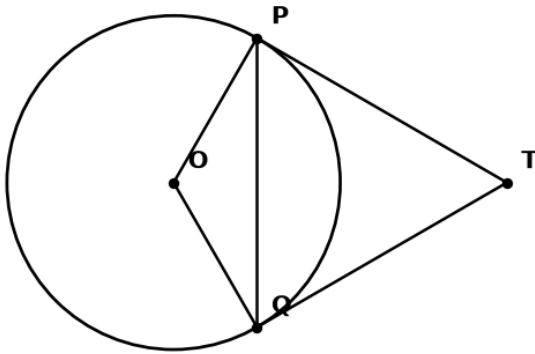


Figure for Q30

OR

All red face cards are removed from a pack of 52 playing cards. A card is then drawn at random from the remaining cards. Find the probability that the card drawn is:

- (i) a face card
 - (ii) a red card
31. A solid is in the shape of a cone standing on a hemisphere with both their radii being equal to 1 cm and the height of the cone is equal to its radius. Find the volume of the solid in terms of π . 3

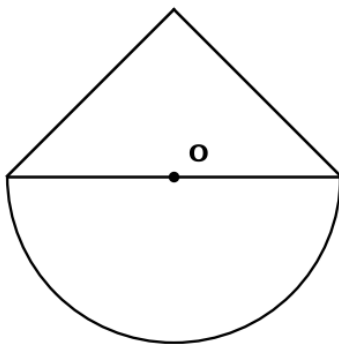


Figure for Q31

SECTION D

This section comprises Long Answer (LA) Type Questions of 5 marks each. $4 \times 5 = 20$

32. Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then the other two sides are divided in the same ratio. 5
33. The sum of the first n terms of an Arithmetic Progression (AP) is given by $S_n = 3n^2 + 5n$. Find the n th term of this AP. Also, find its common difference. 5

34. A solid iron pole consists of a cylinder of height 220 cm and base diameter 24 cm, which is surmounted by another cylinder of height 60 cm and radius 8 cm. Find the mass of the pole, given that 1 cm^3 of iron has approximately 8 g mass. (Use $\pi = 3.14$) 5

OR

From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45° . Determine the height of the tower.

35. The following table gives the production yield per hectare of wheat of 100 farms of a village. 5

Production Yield (in kg/ha)	50-55	55-60	60-65	65-70	70-75	75-80
Number of Farms	2	8	12	24	38	16

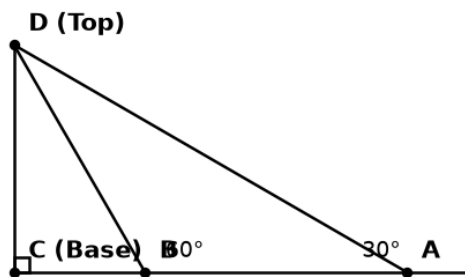
OR

Find the coordinates of the points which divide the line segment joining A(-2, 2) and B(2, 8) into four equal parts.

SECTION E

This section comprises 3 Case Study Based Questions of 4 marks each. $3 \times 4 = 12$

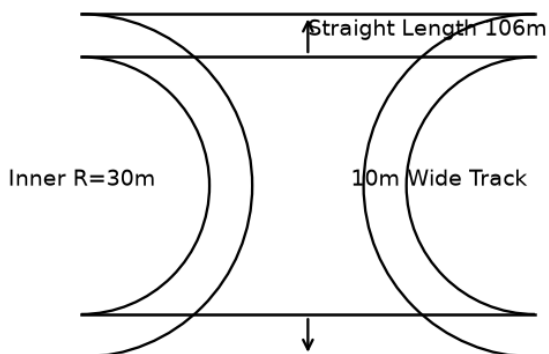
36. A boat is approaching a lighthouse situated at a height of 75 m above sea level. From a point A on the boat, the angle of elevation of the top of the lighthouse is observed to be 30° . After some time, the boat moves towards the lighthouse, and from point B, the angle of elevation of the top of the lighthouse is observed to be 60° . Assume the points A, B, and the base of the lighthouse are collinear and the observation points are at sea level. 4



Observation from a boat approaching a lighthouse.

- (i) What is the distance of the boat from the base of the lighthouse when the angle of elevation is 60° ? [1]
- (ii) What is the distance of the boat from the base of the lighthouse when the angle of elevation is 30° ? [1]
- (iii) Find the distance covered by the boat during the period of observation. (Use $\sqrt{3} = 1.732$) [2]

37. A running track is in the shape of a rectangle with two semicircular ends. The length of each straight parallel section is 106 m and the distance between the two parallel straight sections (inner width) is 60 m. The track is 10 m wide throughout. 4



A running track with semicircular ends.

- (i) Find the inner perimeter of the track. (Use $\pi = 22/7$) [1]
(ii) Find the outer perimeter of the track. (Use $\pi = 22/7$) [1]
(iii) Find the area of the track. (Use $\pi = 22/7$) [2]
38. A survey was conducted on 100 students of Class X regarding the marks they obtained in a mathematics examination. The data collected is presented in the following frequency distribution table: 4

Marks Obtained	Number of Students (Frequency)
0-20	10
20-40	25
40-60	30
60-80	20
80-100	15

- (i) Identify the modal class from the given distribution. [1]
(ii) Determine the median class of the distribution. [1]
(iii) Calculate the mode of the marks obtained by the students. [2]